

CYBERSECURITY INTERNSHIP REPORT

Week 3: Advanced Security Assessment & Reporting

1. Penetration Testing Results

A series of vulnerability assessments were conducted on the local OWASP Juice Shop instance using both automated tools and manual exploitation techniques.

1.1 Network Reconnaissance (Nmap)

Using Nmap (Zenmap GUI), a service scan was executed on port 3000 to identify the underlying technology stack.

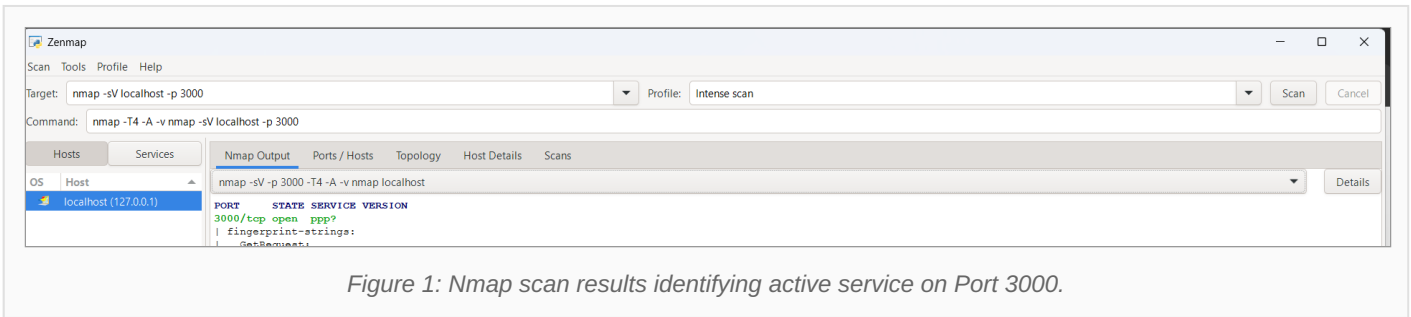


Figure 1: Nmap scan results identifying active service on Port 3000.

1.2 SQL Injection (SQLi) Exploitation

The login interface was tested for improper input handling. By entering a tautology payload, the authentication logic was bypassed.

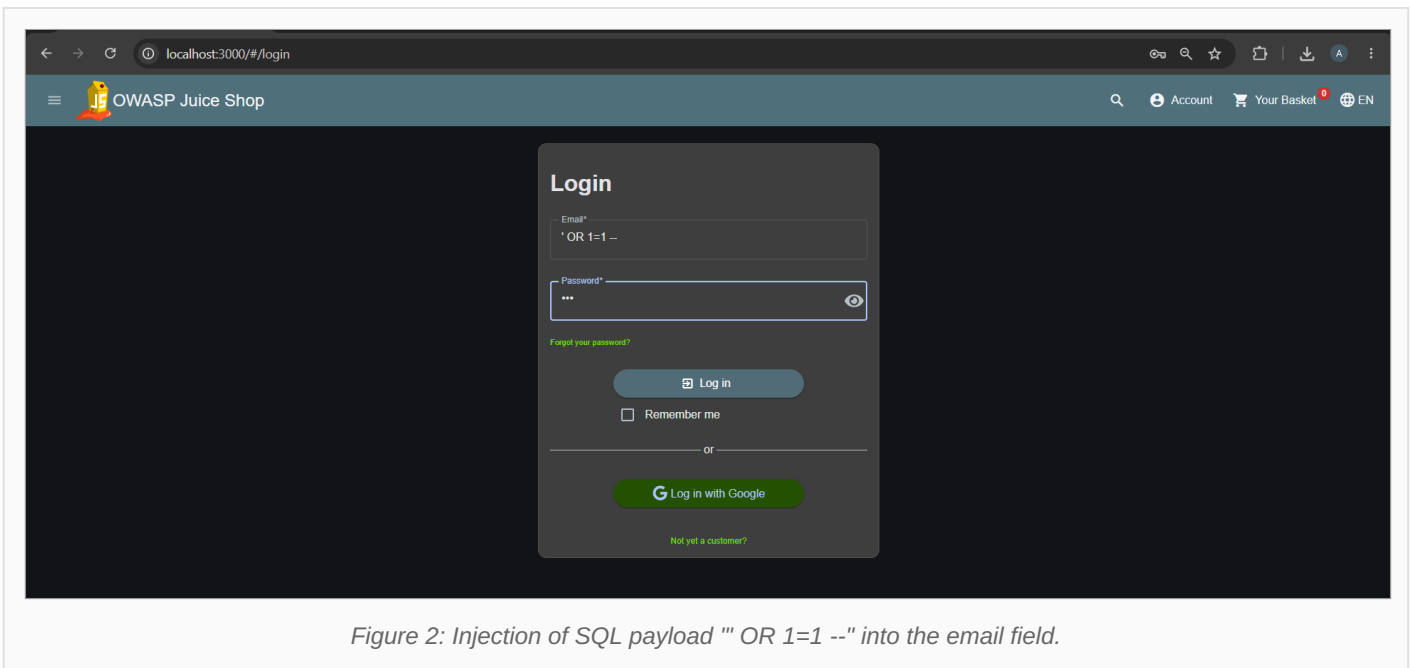


Figure 2: Injection of SQL payload "' OR 1=1 --" into the email field.

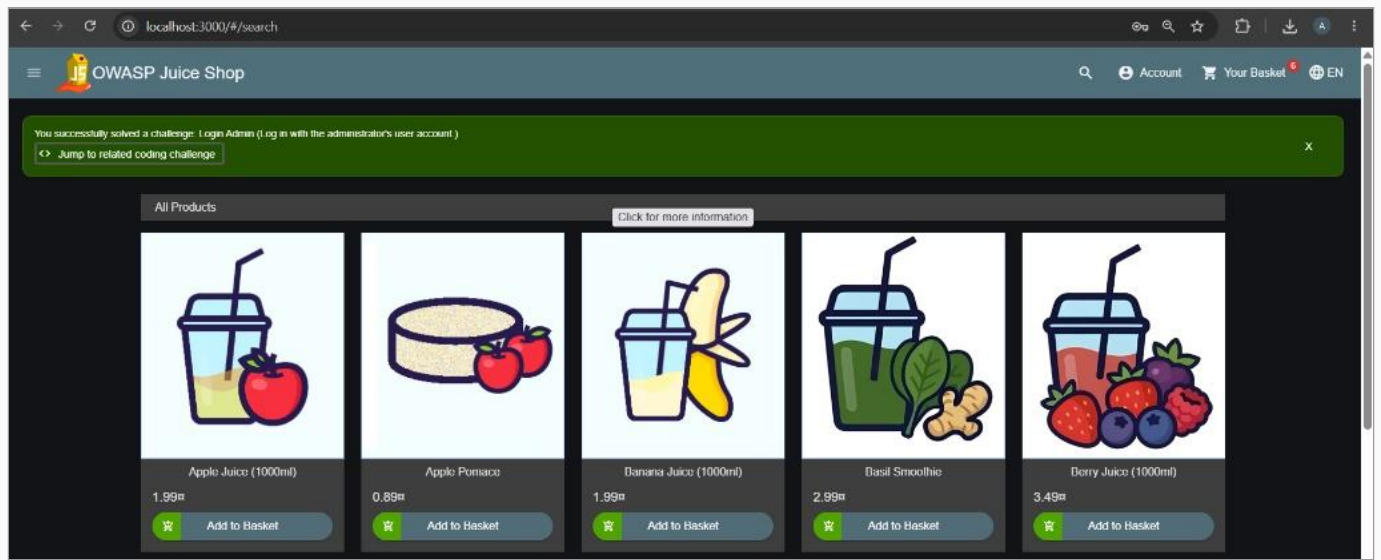


Figure 3: Successful administrative bypass confirmation.

1.3 Cross-Site Scripting (XSS) Testing

A reflected XSS attack was simulated by injecting a script into the search query parameter, confirming the lack of output encoding.

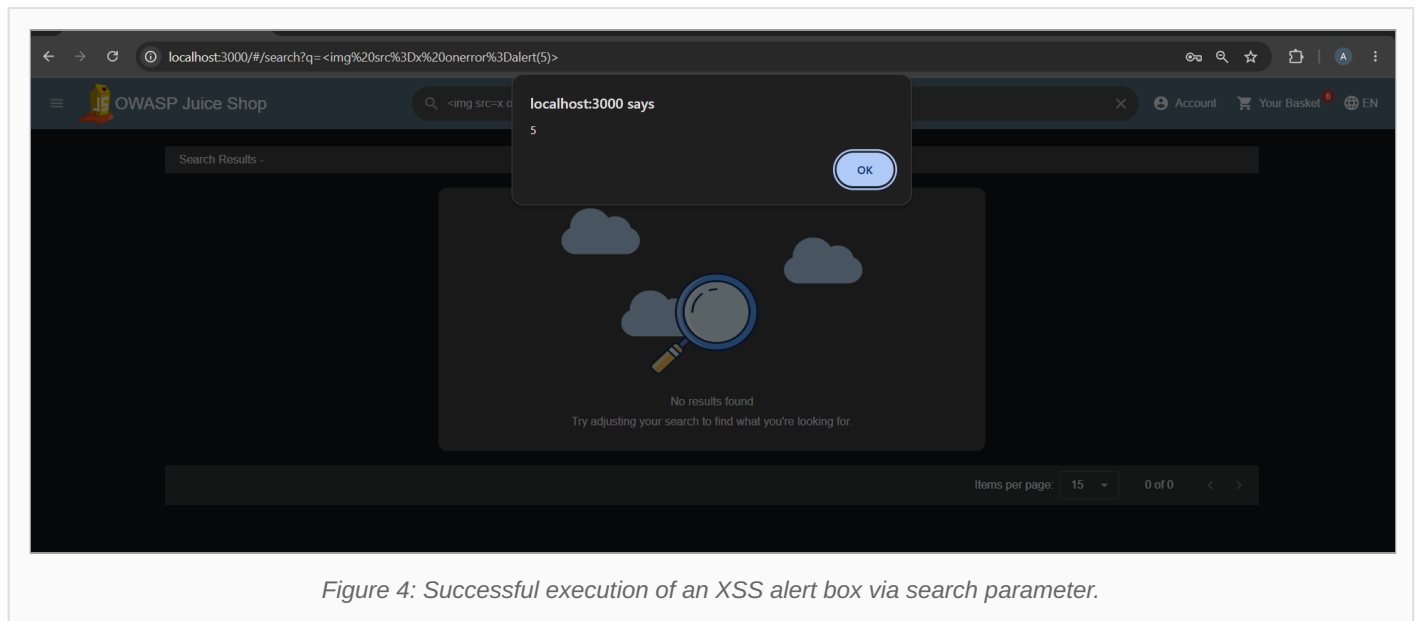


Figure 4: Successful execution of an XSS alert box via search parameter.

2. Application Security Logging

To enhance monitoring, the Winston logging library was integrated. This ensures that system events are recorded for future forensic analysis.

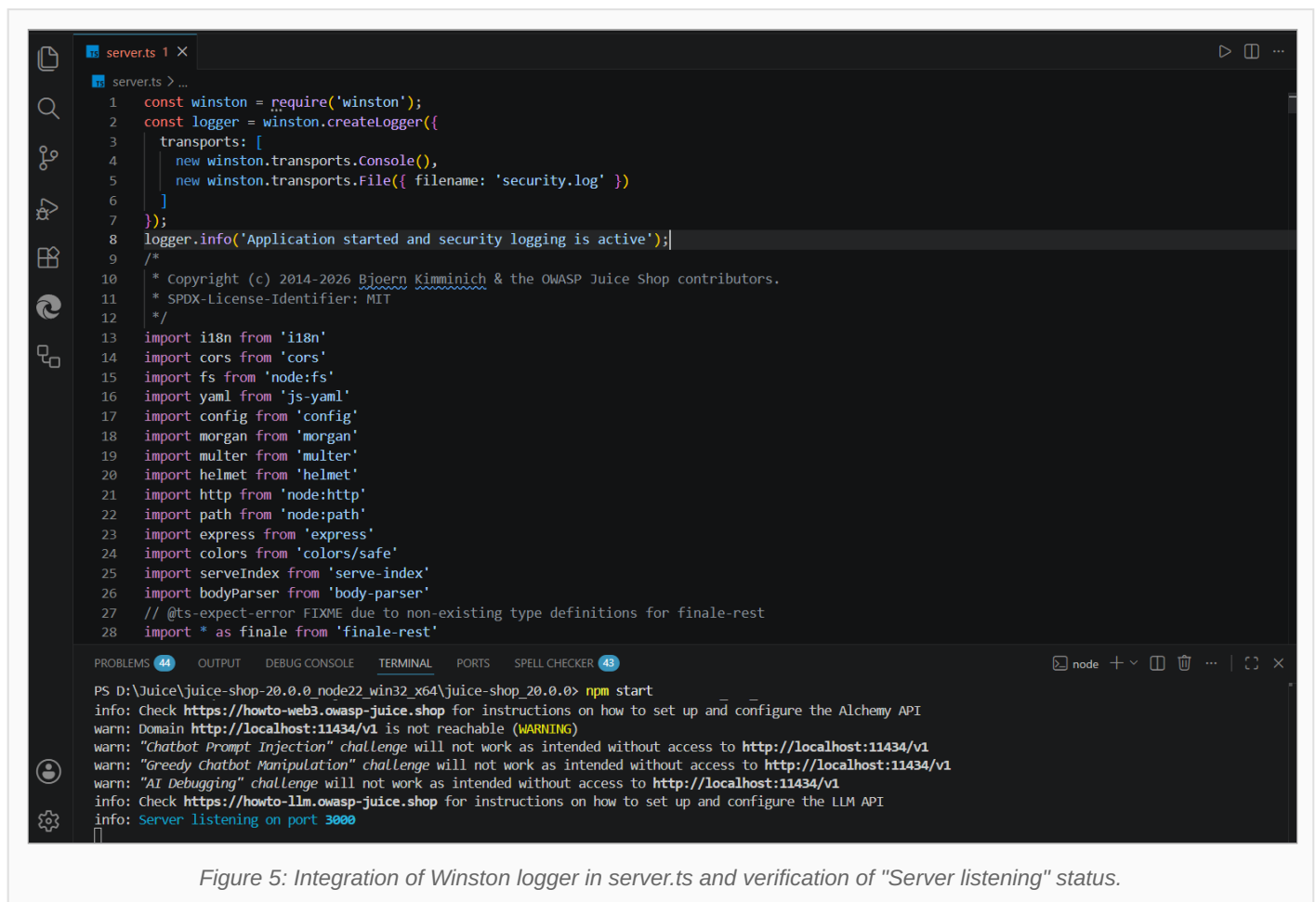


Figure 5: Integration of Winston logger in server.ts and verification of "Server listening" status.

3. Final Security Checklist

Security Best Practice	Status	Observation
Validate all inputs	FAILED	Vulnerable to both SQLi and XSS as demonstrated above.
Use HTTPS for transmission	FAILED	Application currently runs on unencrypted HTTP.
Hash and salt passwords	PASSED	Internal verification shows passwords are stored as bcrypt hashes.

Report Generated for Week 3 Cybersecurity Academic Requirements